

Expected Utility Theory (EUT)

Expected Utility Theory is a decision-making theory developed by **John von Neumann and Oskar Morgenstern**. It explains **how a rational person should make decisions when the outcome is uncertain.**

Instead of choosing the option with the highest monetary value, a person chooses the option that gives the **highest expected utility (satisfaction or happiness)**.

Simple Definition

Expected Utility Theory states that when faced with uncertainty, a rational person chooses the option that provides the highest expected level of satisfaction (utility), not necessarily the highest amount of money.

Key Concepts:

1. Utility

Utility is the **satisfaction or benefit** a person gets from an outcome.

Example:

- ₹100 may be very valuable to a student.
- ₹100 may mean very little to a billionaire.

Both receive the same money, but their **utility is different**.

2. Probability

Every possible outcome has a probability of occurring.

Example:

- Winning a lottery: 10%
- Losing: 90%

3. Expected Utility

Expected Utility is calculated as:

$$EU = \sum P_i \times U_i$$

where:

- **P** = Probability of an outcome
- **U** = Utility of that outcome

Example

Example 1 (Easy)

You have two options.

Option A

Receive ₹500 for sure.

Utility = 50

Expected Utility:

$$EU = 1 \times 50 = 50$$

Option B

50% chance of ₹1200

50% chance of ₹0

Utility:

- ₹1200 → Utility = 90
- ₹0 → Utility = 0

Expected Utility

$$EU = (0.5 \times 90) + (0.5 \times 0)$$

$$= 45$$

Since

Option A = 50

Option B = 45

A rational person chooses **Option A**.

Example 2 (Investment)

Suppose you have ₹1 lakh.

Investment A

Guaranteed return = ₹10,000

Utility = 70

Expected Utility = 70

Investment B

40% chance of earning ₹30,000

60% chance of earning ₹2,000

Suppose

Utility of ₹30,000 = 100

Utility of ₹2,000 = 40

Expected Utility

= (0.40×100)

• (0.60×40)

= $40 + 24$

= **64**

Since

$70 > 64$

Expected Utility Theory says choose **Investment A**.

Assumptions of Expected Utility Theory

1. Decision makers are rational.
2. People know all possible outcomes.
3. Probabilities are known.
4. People prefer higher utility.
5. Preferences are consistent.
6. People maximize expected utility.

Risk Preferences

1. Risk-Averse Person
2. Risk-Neutral Person
3. Risk-Seeking Person

1. Risk-Averse Person

Prefers certainty.

Example:

Guaranteed ₹500

vs

50% chance of ₹1200

Many people choose the guaranteed ₹500.

Examples:

- Buying insurance
- Fixed deposits

2. Risk-Neutral Person

Only cares about expected value.

If two options have the same expected value, either is acceptable.

Example:

- Investors making purely mathematical decisions.

3. Risk-Seeking Person

Prefers risky choices.

Example:

Guaranteed ₹500

vs

50% chance of ₹1500

Many gamblers choose the second option.

Real-Life Applications

Insurance

People buy insurance even though they may lose money overall because the certainty of avoiding a large loss provides higher utility.

Stock Market

Investors choose portfolios that maximize expected utility rather than simply chasing the highest possible return.

Medical Decisions

Doctors and patients compare treatments by considering both:

- Probability of success
- Benefit (utility) of each outcome

Business

Companies evaluate risky projects by considering:

- Possible profits
- Possible losses

- Probability of each outcome
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Government Policy

Governments assess projects such as infrastructure or public health programs by weighing uncertain costs and benefits.

Limitations

- Assumes people always behave rationally.
- Measuring utility is difficult because satisfaction varies across individuals.
- People often make emotional or biased decisions.
- Real-life choices may violate the theory (for example, people sometimes overvalue unlikely events or avoid losses more strongly than the theory predicts).

Expected Utility vs Expected Value

Expected Value

Focuses on money or numerical outcomes.

Assumes ₹100 has the same value to everyone.

Does not account for risk preferences.

Expected Utility

Focuses on satisfaction (utility).

Recognizes that the same amount of money can provide different satisfaction to different people.

Accounts for risk aversion, risk neutrality, and risk seeking.

